

# DIMITRI CHAKMA

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## Summary

Python developer who designed and shipped two production-deployed agentic AI systems end to end, from architecture through Docker and Railway deployment. Built a CBT and mental-health RAG agent with hybrid Neo4j graph and Pinecone vector retrieval, a self-correcting retrieval router, and a crisis-safety gate, alongside a habit-coaching agent using LangGraph, retrieval-augmented memory, and an LLM-as-a-Judge evaluation suite. Backend foundation in FastAPI, PostgreSQL and pgvector, and JWT authentication, backed by Machine Learning, Deep Learning, and NLP specializations from DeepLearning.AI.

## Education

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| <b>United International University</b><br>M.Sc. in Computer Science & Engineering, Major in Data Science (Ongoing)    | 2024 – Present |
| <b>Ahsanullah University of Science &amp; Technology</b><br>B.Sc. in Electrical & Electronic Engineering, CGPA: 3.084 | 2023           |
| <b>Government Rajendra College</b><br>Higher Secondary Certificate (HSC), GPA: 4.42                                   | 2017           |

## Technical Skills

- **Programming Languages:** Python, JavaScript, TypeScript, R
- **Generative AI & Agentic Systems:** LangChain, LangGraph, RAG (Pinecone, pgvector), Knowledge Graphs (Neo4j), LLM Evaluation (Ragas, DeepEval), MCP
- **Backend & APIs:** FastAPI, REST API design, JWT authentication, SQLAlchemy
- **Databases & Tools:** PostgreSQL, pgvector, Neo4j, Pinecone, SQL, SQLite, Git
- **Cloud & DevOps:** Docker, Railway, Streamlit Cloud, LangSmith (tracing), pytest
- **BI & Data Tools:** Microsoft Power BI, Excel
- **Languages:** Bangla, English, Hindi

## Projects

### AI Habit Tracker | Personal project | [GitHub](#)

- Built a habit coaching agent that checks six weeks of history before responding to a missed day, suggesting a smaller version of the habit only for a real recurring pattern.
- Gave the agent long term memory by writing a weekly behavioral summary, embedding it, and storing it in pgvector so advice stays grounded in the user's real past weeks.
- Ran 6 golden coaching scenarios through the real agent, judged by a stronger Claude model, catching two real bugs including an internal tool name leaking into a reply.
- Layered safety as two checks, an input classifier blocking harmful requests before the model runs, plus an output check for leaked internals behind a rate limiting gateway.
- Built a standalone MCP server, kept separate from the production agent, exposing only safe tools like logging and viewing habits while excluding the destructive delete function.
- Deployed an async FastAPI backend on Railway running APScheduler jobs for reminders and summaries, with a Telegram bot interface, Neon Postgres, and LangSmith tracing.
- Stack: Python, FastAPI, LangChain/LangGraph, PostgreSQL/pgvector, Docker, Railway, DeepEval, LangSmith, JWT, MCP.

### CBT & Mental Health Companion (RAG Agent) | Personal project | [GitHub](#) | [Live Demo](#)

- Combined naive RAG and graph RAG under a corrective RAG router that picks vector search, graph search, or both per question, falling back to web search when both are weak.
- Ran the safety check and question preparation in parallel to avoid latency, then fixed a real bug that wrongly flagged a short follow up question as off topic.
- Routed every model call through one rate limited entry point with a shared cache and a daily spend ceiling, then added injection defenses to text pulled from the web fallback.
- Built two evaluation layers, Ragas metrics judged partly by a stronger Claude model, and a custom harness checking graph answers against live Cypher queries and grading crisis gate accuracy.
- Stack: Python, TypeScript, LangGraph, FastAPI, Next.js, Neo4j, Pinecone, Claude (Anthropic), Postgres, Auth.js/OAuth, Ragas, Docker, Railway.

## Research Paper Reading Assistant (ReAct Agent) | Personal project | [GitHub](#) | [Live Demo](#)

- Designed two agent architectures in one project, a ReAct agent choosing its own tool calls for open ended questions, and a supervisor routing once to parallel steps.
- Found and fixed a real Claude API limitation where structured output and citations in one call silently drop the citations, by splitting the work into two separate calls.
- Cut a repeated classification step from 6.8 seconds and 8 cents to 1.7 seconds and zero cost by replacing an LLM judgment call with a typed TypeSafe model.
- Built a real pause into the study guide using LangGraph's interrupt function checkpointed to SQLite, so closing the app mid quiz still resumes at the same locked section.
- Built a full PDF pipeline with PyMuPDF, including column order correction and figure and equation cropping, feeding both the question answering agent and the structured extraction step.
- Added an arXiv search mode that ranks results by recency among topically relevant matches instead of plain keyword frequency, so a user can find papers before ever uploading one.
- Stack: Python, LangChain, LangGraph, Claude (Anthropic), OpenAI, TypeSafe, PyMuPDF, Pydantic, SQLite, Streamlit, Docker, Railway.

## Certifications & Training

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- Complete Generative AI Course with Langchain and Huggingface – Udemy
- Ultimate RAG Bootcamp using Langchain, Langgraph & Langsmith - Udemy
- Data Analysis with Pandas & Python — Udemy
- Microsoft Power BI — Udemy
- Data Analytics — Bohubrihi